

Calculus I Lesson 39

1) The Second Fundamental Theorem of Calculus says that $D_x \left[\int_0^x (t^2 + 1) \cdot dt \right] = \underline{\hspace{2cm}} ?$

2) The Second Fundamental Theorem of Calculus says that $D_x \left[\int_0^x (t^3 + 5) \cdot dt \right] = \underline{\hspace{2cm}} ?$

3) The Second Fundamental Theorem of Calculus says that $D_x \left[\int_0^x (t^2 - 5t + 1) \cdot dt \right] = \underline{\hspace{2cm}} ?$

4) The Second Fundamental Theorem of Calculus says that $D_x \left[\int_1^x (\sqrt{t} + 4) \cdot dt \right] = \underline{\hspace{2cm}} ?$

5) The Second Fundamental Theorem of Calculus says that $D_x \left[\int_1^x \frac{1}{t^2} \cdot dt \right] = \underline{\hspace{2cm}}$

6) a) The Second Fundamental Theorem of Calculus says that $D_x \left[\int_1^x \frac{1}{t} \cdot dt \right] = \underline{\hspace{2cm}}$ for $x > 0$

b) $\int \frac{1}{t} \cdot dt = \int t^{-1} \cdot dt = \underline{\hspace{2cm}} ?$

c) $\int \frac{1}{x} \cdot dx = \int x^{-1} \cdot dx = \underline{\hspace{2cm}} ?$

Definition of the Natural Logarithmic Function: $\ln x = \int_1^x \frac{1}{t} \cdot dt, \quad x > 0.$

7) a) $\ln 1 = \int_1^1 \frac{1}{t} \cdot dt = \underline{\quad ? \quad}$ Note: This is the area from $t = 1$ to $t = 1$.

b) $\ln 2 = \int_1^2 \frac{1}{t} \cdot dt = \underline{\quad ? \quad}$ Hint: Use Area Under Curve Program

c) $\ln 3 = \int_1^3 \frac{1}{t} \cdot dt = \underline{\quad ? \quad}$ Hint: Use Area Under Curve Program

d) $\ln 4 = \int_1^4 \frac{1}{t} \cdot dt = \underline{\quad ? \quad}$ Hint: Use Area Under Curve Program

8) Definition of Natural Log: $\ln x = \int_1^x \frac{1}{t} \cdot dt$

a) $D_x[\ln x] = D_x \left[\int_1^x \frac{1}{t} \cdot dt \right] = \underline{\hspace{2cm}} ?$

Hint: The Second Fundamental Theorem of Calculus

b) $\int \frac{1}{x} dx = \underline{\hspace{2cm}} ?$

9) Evaluate $\int \frac{5+x^2}{x} dx$.

Hint: $\frac{5+x^2}{x} = \frac{5}{x} + \frac{x^2}{x} = 5 \cdot \frac{1}{x} + x$

a) $\int \frac{5}{x} dx = 5 \cdot \int \frac{1}{x} dx = \underline{\hspace{2cm}} ?$

b) $\int x \cdot dx = \underline{\hspace{2cm}} ?$

c) $\int \frac{5+x^2}{x} dx = \underline{\hspace{2cm}} ?$

10) Evaluate $\int \frac{8+x^3}{4x} dx$.

Hint: $\frac{8+x^3}{4x} = \frac{8}{4x} + \frac{x^3}{4x} = 2 \cdot \frac{1}{x} + \frac{1}{4} \cdot x^2$

a) $\int 2 \cdot \frac{1}{x} dx = 2 \cdot \int \frac{1}{x} dx = \underline{\hspace{2cm}} ?$

b) $\int \frac{1}{4} \cdot x^2 \cdot dx = \frac{1}{4} \cdot \int x^2 \cdot dx = \underline{\hspace{2cm}} ?$

c) $\int \frac{8+x^3}{4x} dx = \underline{\hspace{2cm}} ?$

11) Evaluate $\int \frac{x^2 + 5x + 8}{x^2} dx$.

Hint: $\frac{x^2 + 5x + 8}{x^2} = \frac{x^2}{x^2} + \frac{5x}{x^2} + \frac{8}{x^2} = 1 + \frac{5}{x} + 8 \cdot x^{-2}$

a) $\int \frac{5}{x} dx = 5 \cdot \int \frac{1}{x} dx = \underline{\hspace{2cm}} ?$

b) $\int 8 \cdot x^{-2} dx = 8 \cdot \int x^{-2} dx = \underline{\hspace{2cm}} ?$

c) $\int \frac{x^2 + 5x + 8}{x^2} dx = \int \left(1 + \frac{5}{x} + 8 \cdot x^{-2} \right) dx = \underline{\hspace{2cm}} ?$

12) Evaluate $\int \frac{x^3 + 2x^2 + 7x}{x^2} dx$.

Hint: $\frac{x^3 + 2x^2 + 7x}{x^2} = \frac{x^3}{x^2} + \frac{2x^2}{x^2} + \frac{7x}{x^2} = x + 2 + \frac{7}{x}$

a) $\int \frac{7}{x} dx = 7 \cdot \int \frac{1}{x} dx = \underline{\hspace{2cm}} ?$

b) $\int \frac{x^3 + 2x^2 + 7x}{x^2} dx = \int \left(x + 2 + \frac{7}{x} \right) dx = \underline{\hspace{2cm}} ?$

